Reg. No. | | | | | | | | | | [18CS5DS05]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

BIG DATA ANALYTICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Explain Various types of Big Data with an Example	**07**
	b.	How data is being generated? Explain Different V's with an Example	**07**
2.	**a.**	Explain Hadoop Ecosystem and its Components with proper diagram.	**07**
	b.	What is Hadoop File System? List all the components of HDFS and explain any three components.	**07**
3.	**a.**	Describe working principles of the following phases of Map Reduce with one common example. **i.** Map Phase **ii.** Reducer Phase **iii.** Shuffle and Sort Phase	**07**
	b.	What is reducer size and replication rate? Explain with example.	**07**

OR

4.	**a.**	Explain Job Scheduling and Task Scheduling with suitable diagram.	**07**
	b.	Write the java code for MAP and REDUCE of word count problem.	**07**
5.	**a.**	Draw a block diagram and discuss the various frameworks that runs under YARN.	**07**
	b.	Write the difference between the following **i.** Node Manager and Resource Manager **ii.** Hadoop 1.x and Hadoop 2.x	**07**

OR

6.	**a.**	What is the difference between PIG and MAP Reduce also explain some features of PIG.	**07**
	b.	Explain the architecture of PIG.	**07**

7.	**a.**		Explain key components of Hive architecture.	**07**
	b.		Briefly explain the join operations using Hive.	**07**

OR

8.	**a.**	**i.**	What is the default file format to import data using Apache Sqoop? Explain shortly.	**07**
		ii.	Explain any 3 basic commands in Hadoop Sqoop and its uses.	
	b.	**i.**	Explain any 3 Salient Features of Apache Flume.	**07**
		ii.	Compare Sqoop and Flume.	